

KATIE O'CONNOR

Senior Software Engineer

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SUMMARY

Senior software engineer and design-systems architect, 11+ years at Discovery Education, on products reaching 45M students and 4.5M educators across 100 countries. Architected and shipped three successive generations of the company's component library. Also builds AI-native applications end-to-end, from local vector-search retrieval systems to full 3D game architecture, pairing deep design-systems and accessibility expertise with daily AI-assisted engineering workflows.

EXPERIENCE

Discovery Education

Senior Software Engineer

Chicago, IL

2015 – 2026

Design Systems Architecture & Leadership

- Contributed across three generations of Discovery Education's component library: founding team for Comet (2016), build team for successor Nebula (2020), owner and primary contributor for current successor nebula-nuxt (2026). Managed 124 components and utilities across Nebula and nebula-nuxt simultaneously.
- Architected nebula-nuxt, a themed component system letting teams replace legacy UI incrementally while migrating to Nuxt, separating visual-language changes from component rebuilds; ratified as the quarter's formal Planning Interval deliverable following a contested internal architecture debate.
- Led the migration audit and stakeholder roadmap for 46 legacy component replacements; personally built 25 of the 26 components delivered in the new system and released nebula-nuxt for platform-layer consumption. Also built a new playground doc site with searchable documentation (Nuxt Content) and a live theme editor.
- Architected and shipped MegaMenu for a high-stakes Back-to-School (BTS) product deliverable in 3 days, then delivered a refactored production version in another 3; smart-detects when to responsively collapse into a linked drill-down list rather than relying on fixed breakpoints.
- Developed custom Claude skills to support migration: a planning skill that determines the right build approach (themed primitive, wrapped primitive, or fully custom component) for each legacy component, and a release-automation skill running at about \$0.02 per use.

AI-Augmented Engineering & Prompt Strategy

- Uses Claude daily for spec-driven development, producing in-depth, well-researched implementation plans before executing.
- Drove the end-to-end cost of shipping a component—build, doc page, and release—to \$3.05 through skill-based automation.
- Developed model-selection and prompt-caching strategies that reduced daily AI spend by approximately 50%, from \$40–\$50 to \$10–\$20, while preserving output quality; shared the cost-efficiency approach across the team.
- Built a dashboard for human-in-the-loop review and approval of AI-generated metadata, designing the API around the review workflow.
- Co-hosts company's AI Office Hours, and designed and presented a cost-efficient stacked-PR strategy there, structuring PRs in advance to reduce rebasing.

Cross-Team Product Engineering & Adaptability

- Integrated nebula-nuxt into the platform layer with production implementations of Product Bar, Global Nav, Page Shell, Button, Badge, Card, Switch, Row, Modal, and Tooltip; built a live theme switcher and component-comparison playground (legacy Nebula vs. nebula-nuxt side by side) to demonstrate the migration.
- Systems consumed by 12 product teams; partnered directly with teams including Techbook, Search, Assessments, Admin tools, and User Onboarding to architect layouts, interfaces, and components across varied product domains.
- Designed production-ready interfaces directly in code with Claude and existing design-system components when designs were incomplete or unavailable, bridging design and engineering to keep product work moving.
- Advised engineers on theming architecture, component placement, and accessibility best practices; redirected in-progress work into the shared library with proper structure.
- Served as accessibility subject matter expert, performing audits, bug fixes, and code review to enforce accessible implementation across products and shared components.

PROJECTS

Main Character — AI Personal Knowledge & Retrieval System

Python, FastAPI, ChromaDB, Anthropic Claude API

- Architected an AI-powered journal application using local embeddings, an entity graph, multi-

level summaries, and semantic retrieval over personal journal entries.

- Built the retrieval pipeline to ground LLM responses in the user's actual journal history rather than generic model knowledge.

GPS3 (Guinea Pig Simulator 3)

Vue 3 (Composition API), TypeScript, Pinia, THREE.js, Vite

- Solo-developed virtual pet care simulator managing 10 interconnected wellness needs across 2D game view and interactive 3D habitat.
- Designed game's visual language in Claude Design, then converted into a standalone token-based design system and ported back into the game as the single source of styling truth.

SKILLS

AI Engineering: Prompt Engineering, Spec-Driven Development, LLM Cost/Model Strategy, Prompt Caching, RAG, Vector Databases, Anthropic Claude API, Claude Design

Frontend & Architecture: Vue 3, Nuxt, TypeScript, Design Systems, Component Architecture, Theming Architecture, Tailwind CSS, Vite, Accessibility

Backend & Infrastructure: Python, FastAPI, ChromaDB, Node.js, Docker

Other: THREE.js, Pinia, User Experience Patterns, Cross-Team Technical Leadership

EARLIER EXPERIENCE

Additional front-end and production roles, Chicago, IL (2008–2015): Sears Holdings Corporation, Razorfish, AlphaZeta Interactive, Elevation, Hewitt Associates, Bankers Life and Casualty, iPort Media, WGN-TV.